Find and Fix: Illegal Cancellation and Lost Solutions

Directions. Each problem shows work a student handed in. For every one:

- produce the complete, correct solution set on the stated interval, in increasing order;
- then name the error in a sentence or two — say which step is illegal and *which* solutions it destroyed or invented.

1. A student solves $2\sin\theta\cos\theta = \sin\theta$ on $0° \le \theta < 360°$ by dividing both sides by $\sin\theta$, getting $2\cos\theta = 1$, and reports $\theta = 60°, 300°$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Name the error.

Answer: ________________

2. A student solves $4\sin^2\theta - 1 = 0$ on $0^\circ \le \theta < 360^\circ$ by writing $\sin^2\theta = \dfrac{1}{4}$, then $\sin\theta = \dfrac{1}{2}$, and reports $\theta = 30^\circ, 150^\circ$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Name the error.

Answer: ________________

3. A student solves $2\cos^2\theta = \cos\theta$ on $0^\circ \le \theta < 360^\circ$ by dividing both sides by $\cos\theta$, getting $2\cos\theta = 1$, and reports $\theta = 60^\circ, 300^\circ$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Name the error.

Answer: ________________

4. A student solves $\sin\theta + 1 = \cos\theta$ on $0^\circ \le \theta < 360^\circ$ by squaring both sides, simplifying to $2\sin\theta(\sin\theta + 1) = 0$, and reports $\theta = 0^\circ, 180^\circ, 270^\circ$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Name the error.

Answer: ________________

5. A student solves $2\sin 2\theta = 1$ on $0^\circ \le \theta < 360^\circ$ by writing $2\theta = 30^\circ$ or $2\theta = 150^\circ$, and reports $\theta = 15^\circ, 75^\circ$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Name the error.

Answer: ________________

6. A student simplifies $\dfrac{\sin\theta + \sin 2\theta}{\sin\theta}$ by cancelling a $\sin\theta$ from every term, and writes that the expression equals 3 for every θ.

(a) Simplify the expression correctly.

Answer: ____________

(b) Evaluate your simplified expression when θ is one third of π radians.

Answer: ____________

(c) Name the error.

Answer: ____________

7. A student solves $\cos 2\theta = \cos\theta$ on $0° \le \theta < 360°$ using $\cos 2\theta = 2\cos^2\theta - 1$, reaches $(2\cos\theta + 1)(\cos\theta - 1) = 0$, and reports $\theta = 120°, 240°$ — saying that "$\cos\theta = 1$ gives nothing in this interval".

(a) Give the complete solution set.

Answer: ____________ degrees

(b) Name the error.

Answer: ____________

8. A student solves $2\sin\theta\cos 2\theta = \sin\theta$ on $0^\circ \le \theta < 360^\circ$. They divide both sides by $\sin\theta$, then solve $\cos 2\theta = \dfrac{1}{2}$ taking 2θ only between 0° and 360°, and report $\theta = 30^\circ, 150^\circ$.

(a) Give the complete solution set.

Answer: ________________ degrees

(b) Two independent errors were made here. Name both, and say which missing solutions each one cost.

Answer: ________________